

NEWARK-HEATH, OH, USA

WMO: 725229

Lat: 40.023N

Lon: 82.463W

Elev: 269

StdP: 98.13

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 04858

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.1	-13.5	-21.2	0.6	-16.2	-18.1	0.8	-12.4	11.2	2.2	10.2	0.8	2.0	250	0.460

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	32.2	23.2	31.0	22.7	29.7	21.9	24.9	30.0	24.0	28.6	23.3	27.5	3.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.2	18.5	27.5	22.5	17.8	26.6	22.0	17.2	25.9	76.8	30.1	73.3	28.8	70.3	27.4	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	8.0	7.1	-22.2	33.8	4.8	1.5	-25.6	34.9	-28.4	35.8	-31.1	36.6	-34.6	37.7		
(5)			-22.5	26.1	4.7	0.9	-25.8	26.8	-28.6	27.3	-31.2	27.8	-34.6	28.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.2	-2.1	-0.8	4.8	11.3	16.8	21.3	22.9	22.0	18.5	12.0	5.9	1.2		
	DBStd	10.09	7.07	6.46	6.04	4.98	4.60	3.21	2.79	2.72	3.81	4.94	5.00	5.49		
	HDD10.0	1371	378	306	184	44	4	0	0	0	1	34	142	277		
	HDD18.3	3059	633	536	420	214	86	11	2	4	43	205	374	532		
	CDD10.0	1814	3	3	23	83	216	338	400	373	257	96	18	4		
	CDD18.3	460	0	0	1	3	38	99	143	118	49	9	0	0		
	CDH23.3	4271	0	1	8	63	389	898	1256	1034	514	106	2	0		
	CDH26.7	1329	0	0	1	6	98	287	432	319	162	24	0	0		
(14) Wind	WSAvg	2.6	3.4	3.2	3.1	3.3	2.5	2.3	1.8	1.6	1.7	2.3	2.7	3.0		
(15) Precipitation	PrecAvg	1032	80	59	82	91	106	110	107	90	81	73	72	80		
	PrecMax	1382	214	117	185	147	196	244	256	184	173	142	130	205		
	PrecMin	740	29	24	27	40	28	20	41	35	30	27	26	30		
	PrecStd	143	42	24	36	26	48	54	51	42	34	34	26	38		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.6	18.8	24.2	27.5	31.2	32.9	33.7	33.0	32.7	29.7	22.3	17.7		
		MCWB	14.4	13.8	16.6	17.8	21.4	22.6	24.9	23.5	22.4	21.5	16.1	14.1		
	2%	DB	14.3	14.9	20.9	25.2	29.1	31.3	32.0	31.3	30.7	26.2	19.1	14.7		
		MCWB	12.3	11.3	14.3	16.3	20.3	22.1	23.9	23.2	22.1	18.7	13.8	12.1		
	5%	DB	11.3	12.1	17.6	23.0	27.6	29.9	30.7	29.9	28.4	23.6	17.1	12.3		
		MCWB	9.3	9.1	12.4	15.2	19.9	21.4	23.1	22.5	20.9	17.1	12.7	10.0		
	10%	DB	7.9	8.9	15.0	20.9	25.7	28.2	29.2	28.4	26.4	21.1	14.8	9.8		
		MCWB	5.6	6.1	11.1	13.9	19.0	20.9	22.3	21.9	19.7	16.1	11.0	7.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.8	15.1	17.3	19.7	23.0	24.7	26.2	25.7	24.3	22.2	17.5	15.0		
		MCDB	17.0	18.5	22.6	25.8	28.8	30.3	31.7	30.5	30.1	28.1	21.5	17.1		
	2%	WB	12.6	12.3	15.2	17.6	21.8	23.6	25.1	24.7	23.1	20.3	15.2	12.4		
		MCDB	14.3	14.4	19.0	22.7	27.1	28.8	30.5	29.2	28.0	24.1	17.7	14.1		
	5%	WB	9.6	9.1	13.6	16.3	20.7	22.9	24.2	23.8	22.1	18.4	13.4	10.2		
		MCDB	11.3	11.7	16.6	21.2	25.8	27.8	28.9	27.8	26.4	21.9	16.3	12.1		
	10%	WB	5.7	5.8	11.5	15.0	19.7	22.0	23.4	22.9	21.0	16.7	11.5	7.6		
		MCDB	7.6	8.7	14.7	19.4	24.3	26.7	27.5	26.4	24.7	20.3	14.4	9.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.7	9.5	10.7	12.5	12.0	11.6	11.1	11.8	12.9	12.3	10.7	8.0		
		MCDBR	11.0	13.7	15.0	16.2	14.5	13.7	12.6	13.3	15.1	16.0	14.4	11.5		
	5% WB	MCWBR	9.5	11.0	9.4	9.0	7.0	6.0	5.7	6.1	7.0	8.6	9.9	9.8		
		MCWBR	10.4	12.7	13.0	13.0	12.3	11.3	10.9	10.7	12.2	13.2	11.7	10.8		
(40) Clear-Sky Solar Irradiance	taub		0.309	0.329	0.355	0.390	0.430	0.451	0.462	0.450	0.394	0.364	0.326	0.305		
	taud		2.437	2.359	2.359	2.282	2.200	2.181	2.165	2.183	2.319	2.394	2.470	2.472		
	Ebn at Noon		856	885	898	884	852	831	818	817	844	833	829	832		
	Edh at Noon		80	100	112	129	143	146	147	141	115	96	78	72		
(44) All-Sky Solar Radiation	RadAvg		1.60	2.43	3.46	4.68	5.46	5.97	5.82	5.40	4.40	3.02	2.04	1.37		
	RadStd		0.16	0.33	0.32	0.43	0.46	0.42	0.33	0.29	0.40	0.27	0.28	0.16		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (42 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		+0.41	N/A	N/A	N/A	N/A	N/A	N/A	+124	+68

Nomenclature: See separate page